

Aman Singh

Junior Network Engineering & Automation

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OBJECTIVE

Specialising in network automation and data center architecture building spine-leaf fabrics, SDN controllers, and automated lab environments from scratch. Goal: work at the intersection of infrastructure and intelligence, designing networks that are faster, smarter, and more resilient, contributing to systems that power businesses, connect communities, and move society forward. Actively pursuing CCNA certification and seeking a team where I can grow fast and help build something that matters.

TECHNICAL SKILLS

Protocols	BGP (eBGP/iBGP), OSPF, VXLAN, EVPN, OpenFlow, ECMP, STP
Automation	Ansible, Containerlab, FRRouting (FRR), Infrastructure as Code (YAML)
SDN / Virt	Mininet, Open vSwitch, Faucet Controller, Docker, Docker Compose
Observability	Prometheus, Grafana, cAdvisor
Programming	Python (network scripting & automation), Bash / Shell
OS / Tools	Kali Linux, Ubuntu, Git, GitHub, VS Code

PROJECTS

► Network Lab Orchestrator

github.com/MrWater00/network-lab-orchestrator

Python · Ansible · Containerlab · Prometheus · Grafana · Docker

Automated IaC platform deploying, configuring & monitoring full network topologies routers, switches, hosts from a single YAML file via Ansible + FRRouting.

Integrated Prometheus + Grafana + cAdvisor observability stack for real-time CPU, memory & network I/O dashboards across all lab containers.

► Spine-Leaf Data Center Fabric

github.com/MrWater00/spine-leaf-fabric

Shell · Containerlab · FRRouting · eBGP · VXLAN · BGP EVPN

Production-grade 2-spine/2-leaf Clos fabric (same architecture as AWS & Google Cloud) with eBGP underlay, ECMP load balancing, and VXLAN/BGP EVPN overlay.

Eliminated MAC flooding and achieved clean L2 extension across L3 fabric using BGP EVPN control plane.

► SDN Load Balancer

github.com/MrWater00/sdn-load-balancer

Python · Mininet · Open vSwitch · Faucet · OpenFlow 1.3

Round-robin load balancer distributing HTTP traffic from a single VIP across 3 backends using programmatic OpenFlow rules with full IP/MAC rewrite at switch level.

Debugged Python 3.13 compatibility migrated from Ryu to Faucet, resolved all flow rule conflicts and ARP challenges independently.

CERTIFICATIONS & EDUCATION

Cisco CCNA · Self-studying · In Progress

BGP, OSPF, VLANs, STP, ACLs, NAT, network security reinforced daily with live Containerlab labs.

BCA Data Science & AI

SRMU, Lucknow · Completed May 2026 · Result Awaited

ADDITIONAL WORK

dns-labs, bgp-labs, shadowScan (Go), gopherShell (Go), HoneyPot (Python), ai-chatbot & more → github.com/MrWater00